INCH-POUND

MIL-DTL-46550D(AR)
11 September 2011
SUPERSEDING
MIL-C-46550C(AR)
w/AMENDMENT 1
14 July 1989

DETAIL SPECIFICATION

CHARGE, DEMOLITION, SHAPED, 40 LB.
M3A1, METAL PARTS ASSEMBLY

This specification is approved for use by U.S. Army Armament Research, Development and Engineering Center (ARDEC) and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the parts and assemblies for one type of demolition kit designated as Charge, Demolition, Shaped, 40 lb., M3A1.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those specified in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to : Commander ARDEC, ATTN: RDAR-QES-E, Picatinny Arsenal, New Jersey 07806-5000 or emailed to aredecstdzn@conus.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.daps.dla.mil.

AMSC N/A FSC 1375
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DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1168	-	Ammunition Lot Numbering and Ammunition Data Card
MIL-STD-1916	-	DOD Preferred Method for Acceptance of Product

(Copies of these documents are available online at https://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue Bldg. 4D, Philadelphia, PA 19111-5-94.)

2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation (see 6.2).

ARDEC Drawings

8858382	-	Charge, Demolition, Shaped, 40 lb., M3A1, Metal Parts Assembly
8858394	-	Charge, Demolition, Shaped, 40 lb., M3A1, Closing Cap Assembly
8858401	-	Stand-off Frame Assembly

(Copies of these drawings are available from US Army ARDEC, AMSRD-AAR-AIS-TD, Picatinny, NJ 07806-5000, or online from Drawing-Request@us.army.mil.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supercedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First Article. When specified, a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Material. Materials, parts and assemblies shall comply with all requirements specified on the applicable drawings and referenced specifications.

3.3 Assemblies. The assemblies shall comply with all requirements specified on applicable drawings and with all requirements specified in the applicable specifications.

3.4 Welds of stand-off frame assembly. The weld of each leg of the stand-off frame assembly shall withstand a free fall of not less than four (4) feet onto a steel or concrete surface, with the leg striking the surface at approximately thirty (30) degrees from a vertical position,

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without evidence of weld failure, either cracking, separating, or pulling apart. The test shall be performed as specified in 4.3.3.1.

3.5 Waterproofness of closing cap assembly and metal parts assembly. The closing cap assembly and the metal parts assembly shall not exhibit leakage when pressurized 2 to 3 psi internally with air for 10 seconds when tested as specified in 4.3.3.2.

3.6 Fit of closing cap assembly and metal parts assembly. The closing cap assembly shall require a force fit to be joined to the metal parts assembly as specified in 4.3.3.3.

3.7 Workmanship. All parts and assemblies shall be accomplished in a thorough workmanlike manner. They shall be clean and free of cracks, dents, burrs, sharp edges, surface defects, dirt, grease, weld splatter, rust, corrosion, and other defects and foreign matter. The cleaning method used shall not be injurious to any part or assembly nor shall the parts be contaminated by the cleaning agent. Surface coatings shall be continuous. All required markings shall be neat and sharply defined.

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4. VERIFICATION

Table I. REQUIREMENTS/VERIFICATION CROSS REFERENCE MATRIX

Method of Verification	Classes of Verification
1- Analysis 2- Demonstration (end item test) 3- Examination 4- Test	A - First Article B - Conformance

Section 3 Requirements	Description	Verification Method				Verification Class		Section 4
		1	2	3	4	A	B	
3.1	First Article	X		X	X	X		4.2
3.2	Material	X		X	X	X	X	4.3
3.3	Assemblies			X	X	X	X	4.3
3.4	Welds of stand-off			X	X	X	X	4.3
3.5	Waterproofness of assemblies			X	X	X	X	4.3
3.6	Fit of assemblies			X	X	X	X	4.3
3.7	Workmanship	X		X	X	X	X	4.3

4.1 Classification of verification. The verification requirements specified herein are classified as follows:

a. First article inspection (see 4.2)

b. Conformance inspection (see 4.3)

4.2 First article inspection.

4.2.1 Submission. When specified, a first article sample shall be submitted for evaluation in accordance with the provisions of 4.2.2 and Table II. The first article sample shall consist of five (5) complete sets of parts and sub-assemblies (i.e. 5 each of every component and sub-assembly), which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials including packaging and packing shall be obtained from the same source of supply as will be used in regular production. A sample containing known defects will not be submitted.

4.2.2 Inspections to be performed. As determined by the Government, the first article samples may be subjected to any or all of the examinations and tests specified in the specification and inspected for compliance with any or all of the requirements of this specification and applicable drawings, including all unlisted characteristics.

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4.2.2.1 Rejection. The first article sample shall be rejected if any of the inspections/tests fail to meet specification or drawing requirements. The Government reserves the right to terminate inspection or test upon failure of any sample to comply with any requirement.

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Table II. FIRST ARTICLE INSPECTION

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE Charge, Demolition, Shaped, 40 lb., M3A1		SHEET 1 OF 1		DRAWING NUMBER (See below) NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	VERIFICATION LEVEL	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE/ INSPECTION METHOD
	Material	5	100%	3.2	4.3
	Parts and sub-assemblies	5	100%	3.3	4.3
	Welds of stand-off frame assembly	5	100%	3.4	4.3.3.1
	Waterproofness of closing cap assembly and metal parts assembly	5	100%	3.5	4.3.3.2
	Fit of closing cap assembly and metal parts assembly	5	100%	3.6	4.3.3.3

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4.3 Conformance Inspection.

4.3.1 Inspection lot formation. The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-1916. Changes to the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot shall contain assemblies from not more than one lot interfix number from one manufacturer.

4.3.2 Examination. Unless otherwise specified in the Conformance Inspection, sampling plans for major and minor defects shall be in accordance with MIL-STD-1916, Verification Level IV for Major defects and Verification Level II for Minor defects.

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Conformance Inspection

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.3.2.1	TITLE Well detonator		SHEET 1 OF 1		DRAWING NUMBER 8858399 NEXT HIGHER ASSEMBLY 8858397
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	VERIFICATION LEVEL	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE/ INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter and length of cavity		IV	3.2	Gage
102	Outside diameter of body, maximum (max.)		IV	3.2	Gage
103	Wall thickness, minimum (min.)		IV	3.2	Gage
Minor					
201	Diameter of flange, max.		II	3.2	Gage
202	Thickness of flange, max.		II	3.2	Gage
203	Length of closed end, max.		II	3.2	Gage
204	Evidence of poor workmanship		II	3.6	Visual

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Conformance Inspection

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.3.2.2	TITLE Adaptor, well, detonator		SHEET 1 OF 1		DRAWING NUMBER 8858398 NEXT HIGHER ASSEMBLY 8858397
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	VERIFICATION LEVEL	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE/ INSPECTION METHOD
Critical	None Defined				
Major					
101	Pitch diameter of thread, min.		IV	3.2	Gage
102	Minor diameter of thread, max.		IV	3.2	Gage
Minor					
201	Overall length, min.		II	3.2	Gage
202	Evidence of poor workmanship		II	3.6	Visual

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.3.2.3	TITLE Detonator well assembly		SHEET 1 OF 1		DRAWING NUMBER 8858397 NEXT HIGHER ASSEMBLY 8858394
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	VERIFICATION LEVEL	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE/ INSPECTION METHOD
Critical	None Defined				
Major					
101	Concentricity of inside diameter with thread		IV	3.3	Gage
Minor					
201	Assembled length of adaptor, min.		II	3.3	Gage
202	Evidence of poor workmanship		II	3.6	Visual

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.3.2.4	TITLE Cap, closing		SHEET 1 OF 1		DRAWING NUMBER 8858396 NEXT HIGHER ASSEMBLY 8858394
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	VERIFICATION LEVEL	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE/ INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Inside depth, min.		II	3.2	Gage
202	Flange improperly formed		II	3.2	Visual
203	Evidence of poor workmanship		II	3.6	Visual

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.3.2.5	TITLE Closing cap assembly		SHEET 1 OF 1		DRAWING NUMBER 8858394 NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	VERIFICATION LEVEL	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE/ INSPECTION METHOD
Critical	None defined				
Major					
101	Concentricity of inside diameter of closing cap to thread		IV	3.3	Gage
102	Dent in thread or detonator cavity		IV	3.3	Visual
103	Tape missing, loose, or mislocated		IV	3.3	Visual
104	Enamel paint or color improper		IV	3.3	Visual
Minor					
201	Length and diameter of cavity inside		II	3.3	Gage
202	Paint in detonator well		II	3.3	Visual
203	Protective coating missing, inadequate, or improper		II	3.3	Visual
204	Evidence of poor workmanship		II	3.6	Visual

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.3.2.6	TITLE Cone		SHEET 1 OF 1		DRAWING NUMBER 8858396 NEXT HIGHER ASSEMBLY 8858382
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	VERIFICATION LEVEL	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE/ INSPECTION METHOD
Critical	None defined				
Major					
101	Concentricity of lip with either datum diameter		IV	3.2	Gage
102	Variation in wall thickness		IV	3.2	Gage
Minor					
201	Diameter at lip		II	3.2	Gage
202	Wall thickness		II	3.2	Gage
203	Total height, max.		II	3.2	Gage
204	Evidence of poor workmanship		II	3.6	Visual

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.3.2.7	TITLE Metal parts assembly, prior to painting		SHEET 1 OF 1		DRAWING NUMBER 8858382
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	VERIFICATION LEVEL	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE/ INSPECTION METHOD
Critical	None defined				
Major					
101	Concentricity of container neck with either datum diameter on cone		IV	3.3	Gage
102	Total length		IV	3.3	Gage
103	Sealer missing or importer at junction of cone and body		IV	3.3	Visual
104	Cone loose or crimp inadequate		IV	3.3	Visual/Manual
105	Bracket weld missing or inadequate		IV	3.3	Visual/Manual
Minor					
201	Outside diameter, max.		II	3.3	Gage
202	Evidence of poor workmanship		II	3.6	Visual

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.3.2.8	TITLE Metal parts assembly		SHEET 1 OF 1		DRAWING NUMBER 8858382
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	VERIFICATION LEVEL	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE/ INSPECTION METHOD
Critical	None defined				
Major					
101	Bare spot on interior surface		IV	3.3	Visual
102	Color coding improper		IV	3.3	Visual
103	Paint improper or inadequate		IV	3.3	Visual
Minor					
201	Carrying strap missing, broken, improperly assembled, or insecure		II	3.3	Visual/Manual
202	Evidence of poor workmanship		II	3.6	Visual

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.3.2.9	TITLE Stand-off frame assembly		SHEET 1 OF 1		DRAWING NUMBER 8858401
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	VERIFICATION LEVEL	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE/ INSPECTION METHOD
Critical	None defined				
Major					
101	Total length, max.		IV	3.3	Gage
102	Thumb screw missing		IV	3.3	Visual
Minor					
201	Color coding improper		II	3.3	Visual
202	Protective coating missing or inadequate		II	3.3	Visual
203	Evidence of poor workmanship		II	3.6	Visual

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4.3.3 Test methods and procedures.

4.3.3.1 Welds of stand-off frame assembly. Unless otherwise noted, a sample of twenty-five (25) stand-off frame assemblies shall be selected from each lot for this test. Verify by test that there are no weld failures. Presence of a weld failure shall reject the lot.

4.3.3.2 Waterproofness of closing cap and metal parts assembly. Unless otherwise noted, a sample of twenty-five (25) closing cap assemblies shall be selected from each lot for this test. The required air pressure shall be applied and the assembly shall then be completely submerged in water. Observation shall be made for a stream or streams of bubbles, of any frequency, indicating leakage in the assembly. Any assembly exhibiting leakage shall be classed defective and removed from the lot.

4.3.3.3 Fit of closing cap assembly and metal parts assembly. Unless otherwise noted, a sample of twenty-five (25) closing cap assemblies and twenty-five (25) metal parts assemblies shall be selected from each lot for this test. The samples of closing cap assemblies and metal parts assemblies shall be segregated into combinations consisting of one unit of each. The closing cap assembly shall be assembled to its metal parts assembly using applicable equipment. Failure to require a force fit shall class a combination defective. Failure of one or more combinations in meeting the requirement shall reject the lot. Any unit failing to meet the requirement shall be classed defective.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2f). When packaging of material is to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1 Intended use. The 40 lb. Shaped Demolition Charge is primarily employed for burring holes in earth, metal, concrete, and paved or unpaved roads

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Provisions for submission of first article samples.

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c. Requirements for Ammunition Lot Numbering, if applicable.

d. Requirement for acceptance inspection equipment (AIE) designs.

e. Certificate of Conformance for each lot or shipment of product.

f. Preservation, packaging, and marking.

6.3 Acceptance inspection equipment. Provisions concerning the AIE and test facilities that will be used to verify the requirements of this specification should be specified in the contract.

6.4 Distribution of ammunition data cards. Distribution of data cards should include the following: Commander RDECOM-ARDEC, U.S. Army, ATTN: AMSRD-AAR-QEM, Picatinny Arsenal, NJ 07806-5000.

6.5 Subject term (key word) listing.

Ammunition
Body
Cap

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Army-AR

Preparing Activity:
Army-AR
(Project 1375-2011-002)

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